George Sideris

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Education

McGill University | Montreal, QC, Canada Aug. 2024 – Fall 2026 (expected)
Master of Science, Mechanical Engineering (Thesis)

- *Thesis:* Learning Autonomous Log Pile Clearing from Point Clouds (Advisor: Prof. Inna Sharf)
- *Coursework:* Applied Robotics; Fundamentals of Computer Vision; Reinforcement Learning; Advanced Topics in Vision and Robotics
- *Teaching Assistantships:* Applied Electronics and Instrumentation (F2024); System Dynamics and Control (W2025); Engineering Economics (F2024, W2025, S2025, F2025, W2026)

McGill University | Montreal, QC, Canada Aug. 2019 – May 2024
Bachelor of Engineering, Mechanical Engineering (Internship Program) – degree conferred 2024

- *Graduate Courses:* Applied Machine Learning; Probability, Statistics and Machine Learning for Mechanical Engineering; Control Systems

Academic Research Experience

Aerospace Mechatronics Lab | McGill University, Canada Aug. 2024 – Sep. 2026
Graduate Research Assistant (Advisor: Prof. Inna Sharf)

- Developed a GPU-parallel Isaac Lab simulation of a hydraulic forestry crane clearing dense log piles, tuned from field measurements.
- Trained a PointNet-based grasp-targeting policy on raw point clouds by behavior cloning from a privileged expert, then fine-tuned it with reinforcement learning; the fine-tuned policy clears 99 of 100 simulated piles.
- Calibrated the simulation against the physical crane and deployed the policies unchanged in twelve field trials, depositing up to 93.8% of log inventory versus 80.4% for a deployed heuristic baseline (M.Sc. thesis; under review at ICRA 2027).

NeuroImaging and Surgical Technologies Lab | McGill University, Canada Aug. 2023 – Sep. 2025
Undergraduate Research Assistant (Advisor: Prof. Louis Collins)

- Led a team of four to build a multi-camera optical tracker for occlusion-robust, sub-millimeter surgical tool tracking; 2nd place, Engine Capstone Design Prize for Entrepreneurship.
- Co-designed and fabricated the camera fixture for the operating room light; co-implemented multi-camera calibration and ArUco-based tracking.
- Developed an adaptive Kalman-filter fusion strategy with online per-camera noise estimation; released hardware, software, and a multi-view video dataset as an open benchmark (submitted to *Medical Image Analysis*).

Aerospace Mechatronics Lab | McGill University, Canada May 2023 – Aug. 2023
Undergraduate Research Assistant (Advisor: Prof. Inna Sharf)

- Created datasets and trained a CNN for grapple and log segmentation, grapple-opening prediction, and log counting; integrated a draw-wire sensor on the grapple for experimental validation.
- Designed an experimental setup emulating a log-loader manipulator for anti-sway trajectory-planner validation (published at ICRA 2025).

Industry Experience

FPInnovations | Pointe-Claire, QC, Canada Nov. 2025 – Jun. 2026
Automation Research Intern, Mitacs Accelerate

- Mitacs Accelerate internship on the project "Development of Autonomy for Log-loading Machines in Canadian Mill Yards." Calibrated the simulated sensing and control of a trailer-mounted hydraulic forestry crane against the physical testbed and ran field trials deploying learned grasp-targeting policies against the deployed heuristic baseline for automated log-pile clearing.

Lockheed Martin CES | Montreal, QC, Canada Jan. 2022 – Aug. 2022
Jet Engine MRO Engineering Intern

- Created dataset of 100+ labelled images from defective shop floor parts using stereo-microscope. Drafted AI integration sections for $8M CAD NGen proposal on automated visual inspection of HPT parts. Designed tools for jet engine maintenance tasks using AutoCAD; produced drawings and procured materials for 10+ tools. Dispositioned 12 cases of defects on engine parts, saving over $150K CAD.

Bombardier Aerospace | Montreal, QC, Canada May 2021 – Dec. 2021
Project Engineering Intern

- Tracked stress and materials & processes items from inception to disposition using Power BI. Developed a data tracking tool for managing aircraft design modifications using VBA, reducing weekly workload by 4 person-hours.

Publications & Manuscripts Under Review

- **G. Sideris**, E. Ayoub, N. Lemieux, H. Fernando, I. Sharf, "Learning Grasp Targeting from Point Clouds for Log Pile Clearing on a Hydraulic Crane." *Under review, IEEE International Conference on Robotics and Automation (ICRA), 2027.* Video: `youtu.be/woyDNW_oN7A`
- **G. Sideris**, J. Cree, A. Stirling, M. Ly, É. Léger, D. L. Collins, "LARK: A Low-Cost, Accurate, Occlusion-Resilient, Kalman Filter-Assisted Tracking System for Image-Guided Surgery." *Submitted to Medical Image Analysis.* Preprint available at Research Square, doi: 10.21203/rs.3.rs-7622287/v1. Video: `youtu.be/ApZ8q9DjB-4`
- I. Jebellat, **G. Sideris**, R. Saif and I. Sharf, "Designing Experimental Setup Emulating Log-Loader Manipulator and Implementing Anti-Sway Trajectory Planner," *2025 IEEE International Conference on Robotics and Automation (ICRA)*, Atlanta, GA, USA, 2025, pp. 1488-1494, doi: 10.1109/ICRA55743.2025.11128481.

Community Involvement & Leadership

OPSIS | **Montreal, QC, Canada** May 2024 – May 2026
Co-Founder (McGill Engine TechAccel program)

- Developed a low-cost, occlusion-resilient surgical tool tracking system for minimally invasive guided surgery. Completed McGill Engine's TechAccel experiential learning program. Awarded Ian McLachlin Prize for Entrepreneurship by McGill Engine. Supervised a business intern during summer 2025, sponsored by McGill Engine.

McGill Vertical Flight Society Projects Team | **McGill University, Canada** Aug. 2020 – June 2023
Co-Captain / Captain / Structures Lead

- *Team Co-Captain (Aug. 2022 – June 2023):* Co-led twenty students to design and build a 20 lb, 10 ft blended-wing tail-sitter with hover-to-cruise transition; ran workshops on electronics, Arduino, and composites. 3rd place and Perseverance Award, 2023 VFS DBVF competition.
- *Team Captain (Aug. 2021 – July 2022):* Led eight students to build a 15 lb, 6.5 ft quadrotor with airfoils; designed custom frame parts, Abaqus FEA, and an Arduino-controlled wing-tilt mechanism. Best Drawing Package award.
- *Structures Lead (Aug. 2020 – July 2021):* Led the sub-team that designed and built the team's first custom hex-copter frame in SolidWorks.

McGill Institute for Aerospace Engineering | **McGill University, Canada** June 2021 – June 2022
VP External

- Organized inaugural AeroFair, a job fair with recruiters from 6 aerospace companies, attended by 100+ students. Organized two Python workshops: Intro to Git/APIs and CNNs for Fault Classification.

Canadian Robotics Competition (CRC) Teams | **Montreal/Laval, QC, Canada** Sep. 2016 – May 2019
Captain / Team Member

- Three seasons of competition robotics: Marianopolis College team captain (2018–2019) and member (2017–2018); Laval Senior Academy team member (2016–2017).

Honours & Awards

- *FRQNT Doctoral Research Scholarship (awarded 2026/04)*: $100,000 provincial research scholarship; ranked 1st of 30 within its evaluation committee.
- *FRQNT Master's Training Scholarship (2025/06 – 2026/12)*: $26,667 provincial graduate research scholarship.
- *Ian McLachlin Prize for Entrepreneurship in Engineering (2024/11)*: McGill Engine prize for an entrepreneurial project with business or social impact.
- *Engine Capstone Design Prize, 2nd place (2024/04)*: McGill Engine prize for final-year projects with startup potential.
- *SURE Award (2023/05 – 2023/08)*: $9,000 undergraduate research traineeship.

Academic Service

- **Peer Reviewer**: IEEE International Conference on Robotics and Automation (ICRA 2026); IEEE Robotics and Automation Letters (RA-L 2026).

Technical Skills

- **Programming**: Python (PyTorch, OpenCV, NumPy, Gymnasium), C++, ROS, Linux (Unix Shell), Isaac Sim, Isaac Lab, Matlab, Simulink, VBA, Excel, LaTeX
- **Design**: SolidWorks, Fusion 360, AutoCAD, Abaqus, Mastercam, Cura